The Fin d monomial box bridge

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Abstract

Step 4 of the general-d equal-ratio monomial milestone (Astra #14). The paper's monomial integral over the genuine box $(0,1]^d \subseteq \mathbb{R}^d$ is identified with the recursive `productIntegral` of unit 172 in the $[0,\infty]$-valued setting:

$$\int_{(0,1]^d} \prod_i x_i^{h_i}\, g\Big(\prod_i x_i^{2k_i}\Big) dx = \prod_i \frac{1}{2k_i} \int_{(0,1]^d} \prod_i t_i^{w_i}\, g\Big(\prod_i t_i\Big) dt, \qquad w_i = \frac{h_i+1}{2k_i} - 1,$$

for every measurable g and *without* an equal-ratio hypothesis (`monomialBoxIntegral_eq_weighted`); for constant weights $w_i = \lambda - 1$ the weighted box integral is `productIntegral` (`weightedBoxIntegral_const_eq_productIntegral`). Zero `sorry`/`axiom`.

1 The things formalised

```
def unitBox (d :  ) : Set (Fin d →  ) := Set.pi univ fun _ => Ioc 0 1
theorem lintegral_unitBox_succ (d) (F) (hF : Measurable F) :
      x in unitBox (d + 1), F x
      =   a in Ioc 0 1,   b in unitBox d, F (Fin.cons a b)
def weightedBoxIntegral (d) (w : Fin d →  ) (g :   →  0∞) :  0∞ :=
    t in unitBox d, (  i, ofReal (t i ^ w i)) * g (  i, t i)
def monomialBoxIntegral (d) (h k : Fin d →  ) (g) :  0∞ :=
    x in unitBox d, (  i, ofReal (x i ^ h i)) * g (  i, x i ^ (2 * k i))
theorem weightedBoxIntegral_succ (d) (w) (g) (hg) : weightedBoxIntegral (d+1) w g
    =   a in Ioc 0 1, ofReal (a ^ w 0)
        * weightedBoxIntegral d (Fin.tail w) (fun z => g (a * z))
theorem weightedBoxIntegral_const_eq_productIntegral (l) :   d g, Measurable g →
    weightedBoxIntegral d (fun _ => l - 1) g = productIntegral l d g
theorem monomialBoxIntegral_eq_weighted :   d h k, (  i, 0 < k i) →   g, Measurable g →
    monomialBoxIntegral d h k g = ofReal (  i, 1 / (2 * k i))
      * weightedBoxIntegral d (fun i => ((h i :  ) + 1) / (2 * k i) - 1) g
```

2 Mathematics

The first coordinate is peeled with the measure-preserving equivalence $\mathbb{R}^{d+1} \simeq \mathbb{R} \times \mathbb{R}^d$ of `measurePreserving_piFinSuccAbove` at index 0, applied to the product of restricted Lebesgue measures, followed by Tonelli; its inverse is Fin.cons (`Fin.insertNth_zero'`). With $\prod_{i\le d} f_i = f_0 \prod_{i<d} f_{i+1}$ this gives the weighted box integral exactly the recursion defining `productIntegral`. The monomial reduction is an induction on d that peels a coordinate, applies the induction hypothesis to $z \mapsto g(a^{2k_0} z)$ and then the one-dimensional weighted power substitution of unit 170 in the peeled variable.

3 Paper-facing meaning

Combined with units 172 and 173, the equal-ratio monomial box integral with $g(z) = e^{-\beta N z}$ is now expressed exactly as $\prod_i \frac{1}{2k_i} \cdot \frac{1}{(d-1)!} \int_0^1 z^{\lambda-1}(-\log z)^{d-1} e^{-\beta N z}\, dz$, whose asymptotic is known. The assembly into the real-valued theorem $I_d(N) \sim \Gamma(\lambda)\beta^{-\lambda}/((d-1)! \prod_i 2k_i)\, N^{-\lambda}(\log N)^{d-1}$ is unit 175.

4 Lean notes

- **Process gotcha.** A heredoc `cat >` silently clobbered the existing `BoxBridge.lean` of unit 80 (the seabed already had a `BoxBridge`); the duplicate import produced a spurious "build cycle detected". Restored from git and renamed the new file `MonomialBoxBridge.lean`. Always `ls` a proposed new filename first.
- `Fin.insertNth_zero'` needs the dependent family fixed: `change Fin.insertNth (  := fun _ : Fin (d+1) =>  ) 0 y.1 y.2 = _` before `exact`.
- `lintegral_prod` must be given the integrand explicitly (`fun y => F (e.symm y)`); otherwise the pattern is `F   e.symm` and the rewrite fails.
- `simp only [Fin.tail]` cannot unfold an *unapplied* `Fin.tail w`; state the unfolding as an `rfl` equation of functions and rewrite with it.
- `(Fin.cons a b : Fin (d+1) →  ) i` needs the type ascription inside statements, otherwise the dependent family is left as a metavariable.
- `← lintegral_const_mul'` fires on the first matching side; use `conv_rhs` to target the right-hand integral.